

Aarthi R

Full Stack Developer (MERN)

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Professional Summary

Full Stack Developer (MERN) certified by IIT-M Pravartak & GUVI | HCL, with end-to-end ownership of 3 production-grade applications covering frontend, backend, database, and deployment. Proficient in React.js, Node.js, Express.js, MongoDB, and REST API design – the exact stack powering high-traffic automotive platforms. Demonstrated ability to independently build, optimize, and ship features fast – from Stripe payment integration to JWT-secured APIs to MongoDB aggregation pipelines. Strong problem-solving foundation with 250+ coding challenges solved. Actively expanding into TypeScript and Next.js.

Technical Skills

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, React Router, Axios, React Context API
Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcrypt.js, CORS, morgan, Nodemon
Database: MongoDB, Mongoose (Schema Design, Indexing), MySQL
Concepts: JSON, Asynchronous Programming, CRUD Operations, API Integration
Dev Tools: Git, GitHub, Postman, Vercel, Render, Netlify, dotenv
Integrations: Stripe API (Webhooks) Cloudinary, jsPDF, Nodemailer
AI Tools: GitHub Copilot, Claude AI – used actively for code generation, review, and debugging
Learning: TypeScript, Next.js, Redux Toolkit

Projects

- Hotel Booking System | MongoDB · Express.js · React.js · Node.js · JWT · Stripe · Cloudinary · Nodemailer** 
- IIT-M Pravartak GUVI Capstone Project – Scored 10/10 for Architecture, Code Quality & Feature Completeness
- ▶ Architected end-to-end hotel booking platform with 3 room categories, real-time availability, and admin dashboard.
 - ▶ Designed 15+ RESTful API endpoints secured with JWT authentication and bcrypt password hashing, protecting all sensitive routes from unauthorized access.
 - ▶ Integrated Stripe payment gateway with test-mode webhook handling, ensuring reliable transaction tracking and payment confirmation flows.
 - ▶ Improved MongoDB query performance by ~35% through strategic schema indexing on booking history and user collections.
 - ▶ Reduced initial page load by ~40% using React lazy loading and code splitting across route-level components – improving Core Web Vitals.
 - ▶ Implemented Cloudinary image management and Nodemailer email notifications for booking confirmations, signup and login – production-quality UX features.
- Invoice Builder Application | MongoDB · Express.js · React.js · Node.js · jsPDF · React Context API** 
- ▶ Built a complete invoice management system with full CRUD operations, persistent MongoDB storage, and multi-user state management.
 - ▶ Implemented jsPDF-based PDF generation handling dynamic line items up to 50 entries with auto-calculated tax, discounts, and totals.
 - ▶ Managed global state across 8+ components using React Context API – eliminating prop-drilling and maintaining clean data architecture.
- Smart Expense Tracker | MongoDB · Express.js · React.js · Node.js · Recharts · Tailwind CSS** 
- ▶ Developed personal finance dashboard with category-based expense tracking, date filtering, monthly summaries, and interactive charts.
 - ▶ Integrated Recharts for bar and pie chart visualisations with real-time data updates on every CRUD operation.

Education

B.E. Biomedical Engineering Kings Engineering College CGPA: 8.5/10	2019 – 2023 Chennai
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Certificates

- IIT-M Pravartak & GUVI | HCL– Full Stack Development (MERN Stack) with AI tools
- Completed hands-on training in Node.js, Express.js, MongoDB, and REST API development

Achievements & Strengths

- ▶ Perfect 10/10 on IIT-M Pravartak GUVI Capstone – recognised for architecture quality, feature depth, and code standards.
- ▶ Deployed 3 production-grade MERN applications with real-world integrations: Stripe payments, Cloudinary media, JWT auth, PDF generation.
- ▶ Solved 250+ coding problems on GUVI CodeKata – demonstrates algorithmic thinking and consistent problem-solving practice.
- ▶ Career transitioned from Biomedical Engineering to Full Stack Development through rigorous, project-based self-upskilling – demonstrates grit, adaptability, and the fast-learner mindset.
- ▶ Independently managed full product lifecycle on all 3 projects: requirements → design → development → testing → deployment – true ownership mindset.